

A parametric CFD study of computer room air handling bypass in air-cooled data centers

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HIGHLIGHTS

- Experimental validation of a CFD model in a data center test cell with CRAH BP.
- CFD quantified the impact of ignoring temperature non-uniformities by TDM.
- TDM has 4.2% error in predicting optimal cooling power for the baseline data center.
- CFD needed in data centers with shallow plenums, low airflow rate and open aisles.
- Cooling energy savings by CRAH BP method vary depending on the data center design.

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ABSTRACT

Computer room air handling (CRAH) bypass (BP) method is an application that reduces the total fan power in air-cooled data centers by lowering CRAH fan speed. BP fans supply the complementary fraction of server airflow from warmer room air through lower-resistance floor openings into the plenum. Hence, chillers need to operate at lower temperatures less efficiently, which poses an optimization problem. The reduced-order modeling tools in the literature adequately predict fan power. However, they assume well-mixed temperature at rack inlets to deal with this optimization problem and fail to quantify the impact of ignoring temperature non-uniformities. This study includes an experimental verification of computational fluid dynamics (CFD) modeling for CRAH BP method in predicting temperatures in a data center test-cell. The subsequent parametric study uses both CFD and reduced-order modeling tools on a more representative quadrant of a large data center to investigate induced and forced CRAH BP in both enclosed and open aisle configurations as well as comparing various plenum heights, IT load and server airflow rate. Results indicate where reduced-order modeling tools perform reasonably well and where the need for CFD emerges and identify various favorable data center designs and operating conditions for CRAH BP method.

1. Introduction

Power Usage Effectiveness (PUE) is the ratio of total energy consumption of the data center facility to that of the information technology (IT) equipment over a year [1]. PUE is not a true energy efficiency metric due to the limitations, such as penalizing improvements in reducing IT power [2,3]. However, it is still the most popular metric for data center facilities. According to a report, more than 55% of data centers have a PUE value greater than 1.8 [4]. As cooling is responsible for the considerable fraction of the total facility energy consumption, it is apparent that most data centers possess ineffective cooling systems and need effective air and thermal management.

In order to form hot and cold aisles, server racks are usually placed back to back and front to front in open-aisle (OA) air-cooled data centers. Schmidt et al. [5] studied various factors, including ceiling and raised-floor height, and airflow direction affecting the cooling efficiency of air-cooled data centers. The temperature non-uniformity at rack inlets stems from air recirculation between the cold and hot aisle, which is the primary reason for inefficiencies in data centers. According to the American Society of Heating Refrigerating and Air Conditioning Engineers (ASHRAE) [6], the recommended inlet temperature range for various classes of IT equipment is between 18 °C and 27 °C. Therefore, most conventional data center operators try to maintain the inlet air temperature of all servers within the recommended envelope by

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Nomenclature**Abbreviations**

BL	baseline
BF	body force
BP	bypass
CA	cold aisle
CFD	computational fluid dynamics
COP	coefficient of performance
CRAH	computer room air handling/handler
DES	detached Eddy simulation
EA	enclosed aisle
fBP	forced CRAH bypass
fEA	enclosed aisle with forced CRAH bypass
FNM	flow network model
fOA	open aisle with forced CRAH bypass
GDCRL	green data center research lab
HA	hot aisle
HX	heat exchanger
iBP	induced CRAH bypass
iEA	enclosed aisle with induced CRAH bypass
ioA	open aisle with induced CRAH bypass
IT	information technology
LES	large Eddy simulation (LES)
LRN	Low Reynolds Number
MBF	modified body force
NTU	number of transfer units
OA	open aisle
POD	proper orthogonal decomposition
PUE	power usage effectiveness
QDC	quadrant data center
RANS	Reynolds-Averaged Navier Stokes
RMSE	root mean square error
ROM	reduced order model

RSM	Reynolds stress model
TDM	thermodynamic modeling
VTAS	Villanova thermodynamic analysis of systems

Latin symbols

P	pressure, Pa
K	loss coefficient, turbulent kinetic energy
T	temperature, °C
u	uncertainty
UA	thermal conductance, kW/K
V	velocity, m/s
W	power, kW

Greek symbols

Δ	difference
ε	thermal effectiveness, turbulent dissipation rate
ρ	mass density, kg/m ³
Φ	CRAH-to-rack air flow ratio

Subscripts

b	bias
bp	bypass
fan	fan
c	cooling
ch	chiller
chw	chilled water
cr	CRAH
i	inlet
max	maximum
o	outlet
p	precision
r	rack

ASHRAE. In an environment with large temperature non-uniformities, chillers operate inefficiently at lower temperatures and over-cool most of the servers to avoid temperatures outside the recommended envelope.

Applying an enclosed aisle (EA) is the most common method to decrease temperature non-uniformity. Reduced temperature non-uniformity can lead to more free cooling hours as it enables the cooling system to work at a higher temperature. In raised-floor EA data centers, CRAH units should supply the whole required airflow for servers, as well as wasted airflow through the floor and rack leakage paths [7], which can be as high as 50% of the CRAH air flow rate in legacy data centers [8,9]. Tatchell-Evans et al. [7] indicated that reducing rack leakage flow can lead to up to 8.8% decrease in data center energy consumption. Thus, CRAH fans form a significant part of cooling infrastructure power use [10].

CRAH bypass (BP) method by Khalifa and Demetriou [11,12,13], reduces the CRAH airflow and supplies the complementary fraction of the server airflow either by inducing or forcing warm air from the hot aisle (HA) into the plenum through BP fans. The CRAH BP approach leads to a noticeable drop in the total fan power as the airflow resistance that BP fans need to overcome is typically an order of magnitude less than that CRAH fans do (due to various components such as heat exchangers (HX) and filters in the CRAH unit). However, because the cooling system has to operate at a lower temperature inefficiently to keep the rack inlet air temperature below 27 °C, there is a trade-off between the cooling power and fan power.

Fig. 1 displays the schematics of two configurations of the CRAH BP method, including induced BP (iBP) and forced BP (fBP). In the iBP

configuration, iBP fans are under the perforated tiles in the cold aisle (CA). At lower CRAH flow rates, iBP fans compensate for the reduced

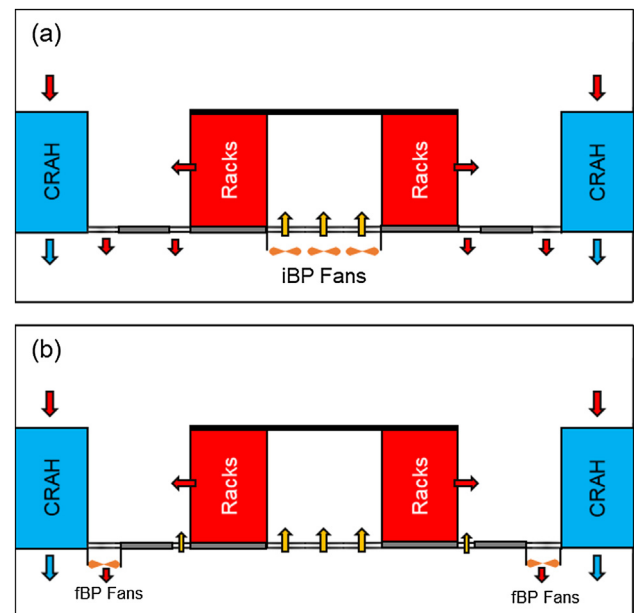


Fig. 1. Schematics of (a) induced and (b) forced BP configurations for air-cooled data centers [14].

airflow by inducing the room air into the plenum through the designated openings over the raised floor. BP openings can be inherent leakage ports (e.g., cable cutouts and tile seams) and intentional openings (e.g., perforated tiles in the HA). The plenum pressure declines at reduced CRAH flow rates and even becomes lower than room pressure when the CRAH flow rate is lower than the needed rack airflow. Negative pressure in the plenum induces the room air into the plenum eliminating the leakage airflow (Fig. 1a). In the fBP configuration, fBP fans are below the perforated tiles in the HA close to the CRAH units. These fans maintain the plenum pressure and leakage at lower CRAH fan speeds. Although the iBP fans provide the entire rack airflow, fBP fans deliver only a fraction of it (Fig. 1b).

Some studies have investigated the CRAH BP method through various modeling tools and experiments. Erden et al. [15,16] experimentally validated a flow network model (FNM) in air-cooled data centers with and without CRAH BP. In their estimation of airflow rate and fan power, there is less than 10% disagreement between FNM and experimental results [16]. The thermodynamic model (TDM) estimating the cooling infrastructure model based on the FNM with the well-mixed temperature assumption has been utilized to find the optimum point in different studies [15,16,17,18]. Erden et al. [17] estimated that the iBP provides a 5% more reduction in cooling power in comparison with fBP. Erden et al. [18] extended their work to build a unified modeling approach for the iBP method and concluded as much as 52% reduction in the cooling power in data centers with indirect airside economizers utilizing the iBP method. Considering that the indirect airside economizer is not the most common free cooling method, the performance of CRAH BP method with other more common methods, such as direct airside economizer [19], indirect evaporative cooler and water-side economizer [20], needs further investigation.

Computational fluid dynamics (CFD) is a powerful tool that allows quantifying the non-uniformities and errors associated with well-mixed temperature assumption in the TDM. Patankar [21] investigated many physical parameters such as plenum height and porosity of perforated tiles with the help of CFD simulation. Despite numerous room-level CFD simulation studies, only a few included the use of fans underneath the perforated tiles. Athavale developed an experimentally validated CFD model for a data center lab with active tiles and found that active tiles by changing plenum pressure influences on the CA airflow rate and amount of leakage from plenum into HA [22]. Erden [23] and Ahmadi and Erden [14] conducted the first CFD-based optimization studies for the CRAH BP method and questioned the validity of the well-mixed temperature assumption in a quadrant data center. However, their findings were not backed up with experimental results.

This study includes the experimental validation of a CFD model against measured temperatures at Syracuse University's Green Data Center Research Laboratory (GDCRL) with induced CRAH BP configuration. The subsequent parametric study based on CFD and TDM

includes both induced and forced CRAH BP methods in a representative quadrant of a 1 MW data center with and without aisle containment to study the impact of plenum height, IT load density, server airflow rate.

2. Methodology

2.1. Modeling

2.1.1. CFD

To determine the temperature and flow field in the air space of the data center, CFD models solve the mass (continuity), momentum and energy equations. The numerical simulation has been executed by using the commercially available CFD software, Fluent 18.1 [24]. The literature contains various CFD modeling studies for the room-level thermal investigation of data centers that focus on the choice of turbulence models [25,26], and perforated tile modeling [25,27,28]. Wilbron et al. compared three turbulent models, including k- ϵ , Reynolds Stress Model (RSM) and Detached Eddy Simulation (DES, a hybrid formulation of Reynolds-Averaged Navier Stokes (RANS) inside the boundary layers and Large Eddy Simulation (LES) somewhere else) [26]. They showed that all three cases have similar deviations from experimental results. Since the k- ϵ model does not predict low-velocity areas in the data centers, and DES is computationally expensive, they suggested the RSM model [26]. However, Phan et al. [25] showed that the computational time for the RSM method is about 70% higher than that of the standard k- ϵ model and greater than any other option where they compared ten different RANS turbulence models. They also considered various options for tile modeling and concluded that body force (BF) [27] and modified body force (MBF) [28] are the best tile modeling tools. For both BF and MBF tile modeling at both medium and high airflow rates, the Low Reynolds Number (LRN) k- ω , SST k- ω , realizable k- ϵ , and standard k- ϵ showed a better agreement with experimental results compared to other turbulence models. Other researchers also confirmed acceptable accuracy through various validation studies utilizing the standard k- ϵ model [22,27,28,29]. Hence, this research employs the standard k- ϵ turbulence model and standard wall function, MBF for perforated tiles, as well as full buoyancy effects [27]. Table 1 shows a summary of CFD model assumptions.

The layouts of the 22 × 38 ft (6.71 × 11.58 m) quadrant of a generic data center (QDC) for both iBP and fBP configurations are in Fig. 2 (top view). The ceiling height above the raised floor and under-floor plenum height are 144 in. (3.66 m) and 36 in. (0.91 m), respectively. The dimensions of each floor tile are 24 × 24 in. (0.61 × 0.61 m).

QDC is made up of three halves and one whole CRAH units, stating 10 CRAH units for the whole data center. The intersection of the symmetry lines represents the center of the data center, where 2 CRAH units exist. Constant velocity inlet boundary conditions in the plenum

Table 1
Summary of CFD model assumption.

Solver Type	–	Pressure-Based
Solver Time	–	Steady State
Turbulence Model	–	Standard k- ϵ
Near-Wall Treatment	–	Standard Wall Functions
Additional Viscous Modeling Options	–	Full Buoyancy Effects
Pressure-Velocity Coupling Scheme	–	SIMPLE
Spatial Discretization	Gradient	Least Squares Cell Based
	Pressure	Second Order
	Momentum	Second Order Upwind
	Turbulent Kinetic Energy (k)	First Order Upwind
	Turbulent Dissipation Rate (ϵ)	First Order Upwind
	Energy	Second Order Upwind
Residuals	Continuity	10^{-4}
	X, Y, Z-velocity	10^{-5}
	Energy	10^{-10}
	k, ϵ	10^{-5}

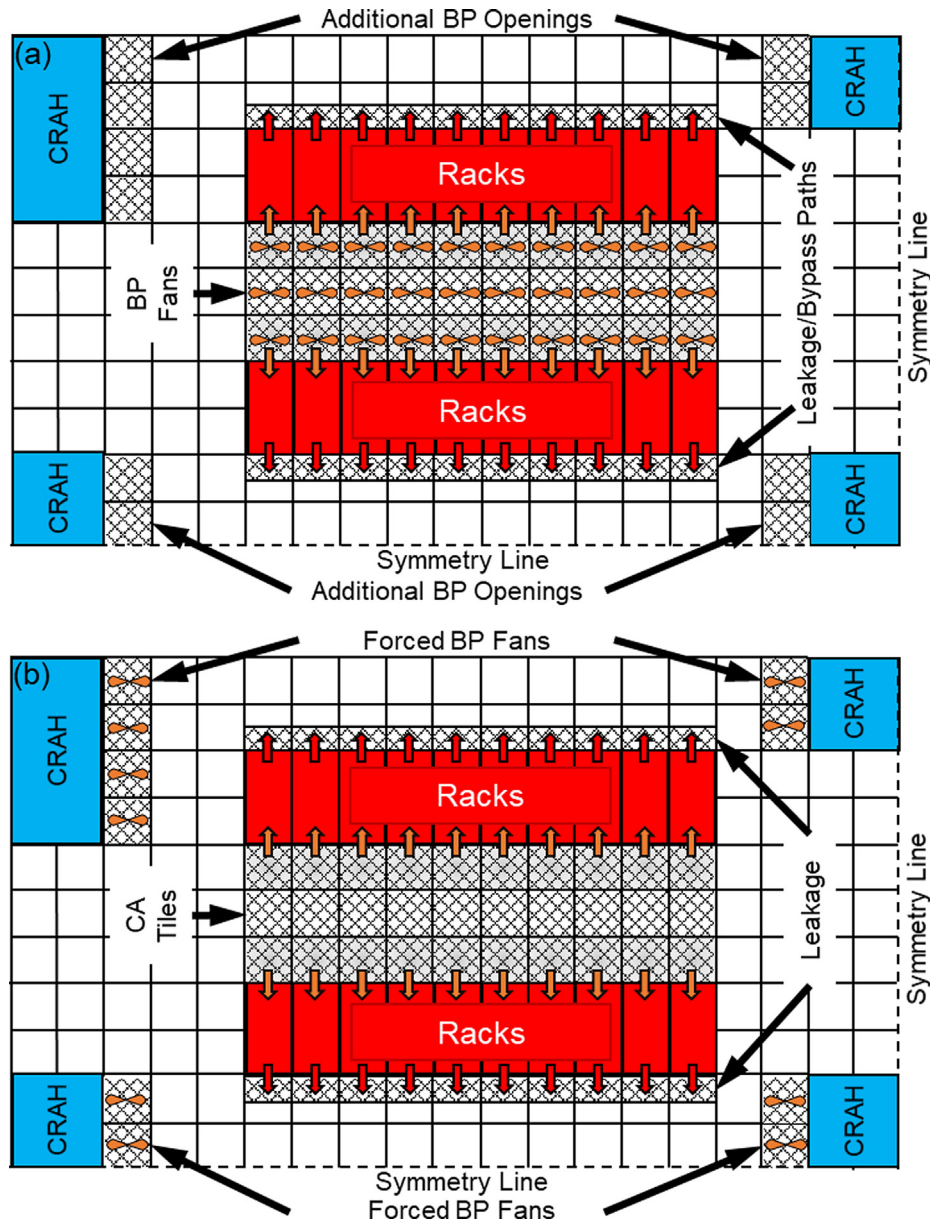


Fig. 2. Top view of the QDC layout for both (a) iBP and (b) fBP configurations [14].

represent CRAH units' exit conditions at a uniform temperature. QDC with uniform IT load distribution has 20 racks in two rows housing 60 blade-type servers. Zhang et al. [30] in CFD simulation did not model inside of server racks and considered them like boxes. They concluded that this kind of simplification does not lead to a significant error with measurements. The recirculation model allows reduced-order modeling of the server without the rack interior modeling and meshing [27]. This model couples inlet and exit surfaces of each server, and to determine the exit boundary condition adds generated IT load heat to the inlet airflow.

The turbulence length for inlet boundaries is considered 7% of the hydraulic diameter of the opening [24]. Turbulence intensity is the ratio of the root-mean-square of the velocity fluctuations to the mean flow velocity, and the amounts greater than 10% are considered high [24]. This study employs the turbulence intensities of 10% and 50% at the exit boundary conditions of servers [27] and CRAH units [13], respectively. Both assumptions are based on experiments conducted at the GDCRL by Abdelmaksoud [27] and Demetriou [31]. Demetriou [31] measured turbulence intensity values of 15–20% and 50–70% in-

line with the CRAH blower exit and between the CRAH blowers in the GDCRL, respectively. He recommended using an average value of 50% for the CRAH exit boundary condition with limited geometrical details. A sensitivity analysis for the range of values mentioned above indicated that there is a negligible effect of CRAH exit turbulence intensity on the optimization results.

The CA in the QDC is 3-tile wide and 10-tile long. Two different openings enable airflow between the room and plenum in the rest of the floor. BP tiles are adjacent to CRAH units, and leakage tiles are at the back of the racks acting as cable cutouts. According to the assumptions by Erden et al. [16], all openings either in the CA or HA have a flow resistance coefficient of $K = 21$. A 4-inch (0.1016 m) thick porous volume assumes these openings with a porosity of 25% and the inertial resistance coefficient along the vertical flow direction. All floor openings incorporate a modified body force (MBF) model [28] to account for the momentum deficit in the absence of the perforated tile geometrical details and subsequent small jets.

Based on whether the iBP or fBP approach is in use, the location of CRAH BP fans changes in QDC. The BP fans in the iBP are in the CA

underneath the perforated tiles (Fig. 2a), and in the fBP, they are in the HA only below the perforated tiles near to CRAH units (Fig. 2b). According to the CFD study by Demetriou and Khalifa, the BP tiles near CRAH units provide more time for mixing and have more uniform temperatures at the CA and rack inlets [13]. As a result, BP fans in fBP configuration, as demonstrated in Fig. 2b, are deliberately beneath perforated tiles next to CRAH units. Similarly, iBP has perforated tiles adjacent to CRAH units to induce room air into the plenum in addition to the inherent cable cutouts. Moreover, this research investigates various configurations, both with and without CA containment (EA and OA).

The BP fan boundary condition, for both configurations, provides a uniform pressure increase across the fan boundary surface, which is parallel to perforated tiles and placed 12 in. (0.3 m) away from the tiles inside the plenum. The CFD model includes separating walls between each BP fan surface. This study investigated the swirl specification of BP fans and led to a negligible impact. Generally, radial velocity is assumed zero, and the reasonable ratio of tangential velocity (V_t) to axial velocity (V_a) is around 1.0 ($V_t/V_a \sim 1$) [32]; however, to see the maximum effect of swirl specification, here ($V_t/V_a \sim 2$) is applied. The resulting difference in cooling power is no more than 3%.

The grid-independence study investigates five different grid sizes, with the total number of cells ranging from about 100 thousand to 20 million to determine the sensitivity of mesh size to the results. The computational region for the QDC consists of hexahedral mesh with element sizes of 1&1, 2&2, 4&4, 4&2, 6&6 in., in which the first and second numbers indicate the grid sizes at the room and plenum, respectively. The cases with uniform 4- and 6-inch mesh sizes require grid adaptation at the plenum walls to maintain y^+ (i.e., $y^+ < 500$) values within recommended limits [33]. Fig. 3 compares the temperature at a particular point at the entrance of the CRAH unit for different mesh sizes. The 4&2 case provides reasonable computation time, and performance as the temperature difference between the finest mesh (1&1) is only 0.3 °C.

2.1.2. TDM

Literature consists of various reduced-order modeling tools as alternatives to CFD modeling with the high computational burden for analyzing the thermodynamics, heat and fluid phenomena in data centers [16,34,35]. Wemhoff et al. [34] introduced the Villanova Thermodynamic Analysis of Systems (VTAS) as a thermodynamic modeling tool to evaluate the cooling methods for data centers with energy inefficiencies by using exergy destruction metric. Fouladi et al. [35] proposed a hybrid model consisted of VTAS and proper orthogonal decomposition-reduced order model (POD-ROM) to compute the exergy destruction of the components (i.e., servers, CRAH, chiller and cooling tower) and airspace, respectively. Erden et al. incorporated FNM to predict the pressures in the data center airspace with the TDM to compute key temperatures with well-mixed properties at various nodes [16]. This study utilizes the explicitly coupled FNM and TDM similar to that of [16] to optimize the cooling power of the QDC at a specific ambient condition.

Fig. 4 shows the schematics of the FNM and TDM. The commercially available modeling tools, AFT Fathom [36] and TRNSYS [37], carry out the FNM and TDM, respectively. For baseline configuration, the flow network of the data center excludes both iBP and fBP fans and the airflow path with the solid green line. CRAH BP configurations, iBP and fBP, have the airflow path with solid green line activated to allow the CRAH BP airflow in addition to the corresponding BP fans (i.e., iBP and fBP). In the FNM, the dominant flow resistances consist of the flow resistance of the CRAH units, racks, perforated tiles, and leakage paths. The following equation estimates the pressure loss due to different resistances,

$$\Delta P = \frac{1}{2} K \rho V^2 \quad (1)$$

where K is the specific loss coefficient, ρ is the air density and V , the air velocity. To calculate fan power, FNM employs the obtained pressure and flow distribution as well as relevant fan characteristics assuming atmospheric pressure at the room in the data center.

TDM resolves well-mixed air temperatures at key points (e.g., plenum, tile, rack exit, CRAH inlet/exit, room, etc.) by applying obtained fan flow rate and power use through FNM as well as setting redline temperature (27 °C) at server inlets. The ε -NTU relations give the thermal effectiveness, ε , of the cross-flow air-to-water CRAH heat exchangers (HX) [38], where NTU is the ratio of the thermal conductance UA to the minimum of the fluid heat capacity ratios. At reduced CRAH airflow rates, UA scales with the 0.8th power of the flow rate according to the change of Nusselt Number for turbulent internal flow with Reynolds Number [38]. Thus, the HX model determines the chilled water temperature (T_{chw}) to be supplied by the chiller model. The chiller model computes the Coefficient of Performance (COP) and chiller power based on a normalized performance map at various chilled water setpoint temperatures [39,40].

2.2. Optimization

For different CRAH-to-rack airflow ratios (Φ) in both iBP and fBP with EA and OA data center, CFD analyses yield temperature and flow fields for QDC. Fig. 5 demonstrates the outline of the TDM-based and CFD-based optimization of the CRAH BP method for only EA [23] and both EA and OA data centers, respectively.

The level of pressurization of the cold aisle has a significant impact on bypass [7]. It is necessary to provide zero-static pressure in rack inlets or inside the aisle containment to ensure that racks are well-provisioned. Well-provisioned racks receive the amount of airflow they need in EA arrangement as they would receive in the OA configuration. The CFD simulations for different Φ values determine the required uniform static pressure increase through the BP fans (ΔP_{bp}). Knowing the pressure and airflow for BP fans, and applying the operating points and efficiency curves lead to the estimate of BP fan power ($W_{fan,bp}$). The obtained $W_{fan,bp}$ is an energy source input in CFD and causes a slight temperature increase over the tiles equipped with BP fans. Provided that the maximum server rack inlet temperature ($T_{r,i-max}$) is under the redline (27 °C), CFD gives the associated CRAH air supply temperature ($T_{cr,o}$). Finally, using the CRAH and chiller components of TDM shown in Fig. 4, as well as the mass-weighted inlet and exit air temperatures at the CRAH unit, the chiller power, W_{ch} , can be determined. The ultimate cooling power is

$$W_c = W_{ch} + W_{fan,cr} + W_{fan,bp} \quad (2)$$

where CRAH fan power ($W_{fan,cr}$) is obtained by the FNM and BP fan power ($W_{fan,bp}$) is based on CFD.

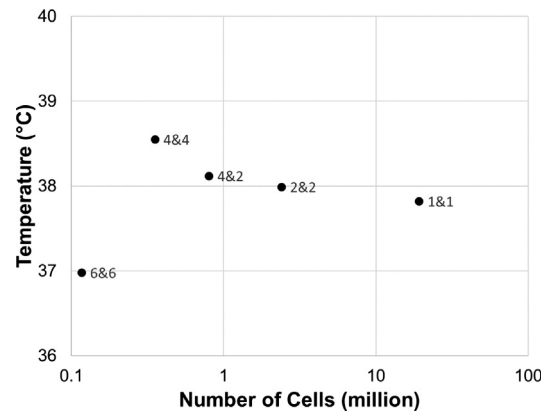


Fig. 3. Temperature variation of a particular point at the entrance of the CRAH unit in different mesh sizes.

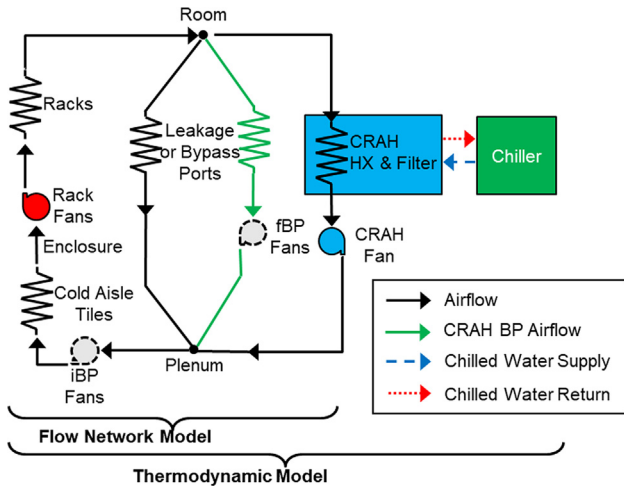


Fig. 4. Schematics of TDM and FNM [14].

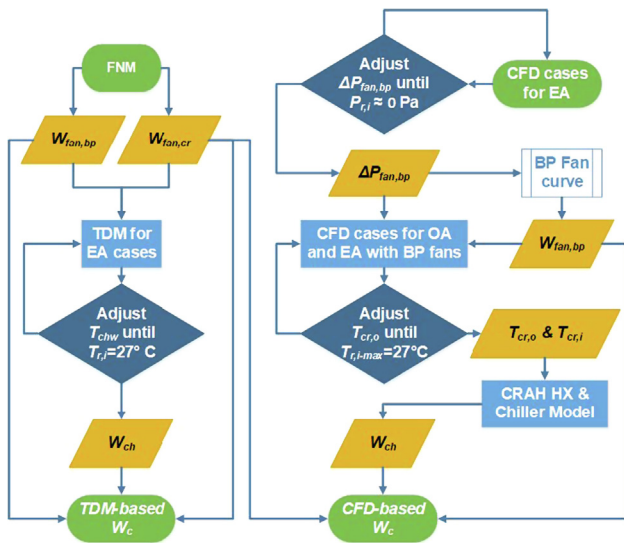


Fig. 5. Outline of TDM-based and CFD-based optimization [23].

2.3. Model assumptions

Table 2 specifies the features of 1 MW IT load data center. The uniform IT load among the server racks increases the temperature of rack airflow by 14.6 °C ($\Delta T_r = 14.6$ °C). The CA in the entire data center includes 120 perforated tiles housing 120 BP fans in iBP configuration (Fig. 2a). All CRAH BP configurations have 40 perforated tiles in the HA near CRAH units, and in fBP configuration, 40 BP fans operate underneath these tiles (Fig. 2b). The leakage paths behind the racks cause roughly 25% of the CRAH airflow rate leakage into the room in the baseline configuration (i.e., $\Phi = 1.33$). Key flow characteristics such as the resistances (CRAH, racks, and tiles) and fan curves (CRAH, server and BP) are based on vendor data and in line with the assumptions in the previous optimization studies [18]. The thermal conductance UA for the cross-flow HX of a CRAH unit is based on vendor catalog data [18]. Chiller cooling capacity is 1500 kW, has a COP of 6.0 and receives condenser cooling water at a temperature of 35 °C.

3. Validation

In order to validate the numerical model, the measured temperatures throughout the server rack inlets and leakages at the GDCRL [16] have been compared with the CFD results. To have a consistent presentation of temperatures in experiments with different CRAH exit air

temperatures ($T_{cr,o}$), a relative temperature of $T - T_{cr,o}$ is defined. This definition relies on the assumption that any change in the $T_{cr,o}$ changes the entire temperature field by the same amount, which is reasonable for environments such as GDCRL with less dominant buoyancy and limited temperature variations (~ 2 °C) in the $T_{cr,o}$ [27]. GDCRL consists of three simulated server racks, one of which was not operational. Two active racks generate a total power of 29.4 kW each with a total airflow rate of 1.151 m³/s (2440 CFM), leading to a temperature increase of ~ 10.6 °C. One of the two CRAH units with a flow capacity of 4.153 m³/s (8800 CFM) provides servers required airflow through the underfloor plenum. The use of a single CRAH unit not only is sufficient for two active server racks but also decreases the experimental uncertainties. The plenum is approximately 1 m deep. CA consists of two rows of three perforated tiles. Two perforated tiles in the HA provide 25% of the CRAH airflow as leakage from the plenum into the room in the baseline case [16]. Fig. 6 demonstrates the layout and geometry of the experimental facility.

Four thermocouple frames (each with nine thermocouples) remain at the top and bottom simulated chassis of R1 and R3 to monitor the server inlet air temperatures. The insulated portable CA containment separates the HA and CA. Three commercially available fans installed below the six perforated tiles in CA, with around 0.5 m distance from plenum, represent the BP fans of the iBP configuration [16].

An uncertainty investigation quantifies the systematic (or bias) and random (or precision) error of temperatures over the server inlets and in the middle of the leakage tiles for the experimental data. The bias error (u_b) consists of calibration and data acquisition error, error due to the uncertainty in the CRAH exit temperature and, the potential location error because of misaligned thermocouple frames [41]. The precision error (u_p) assumes two standard deviations (2σ) (95% confidence interval) for temperature measurements [41]. The total uncertainty (u) is as follows,

$$u = \sqrt{u_b^2 + u_p^2} \quad (3)$$

The operating CRAH unit in GDCRL has two small exit areas associated with the blowers and an angled vane to conduct the airflow towards the CA. Previous studies [14,23] used the large opening at the entire floor area of the CRAH unit as the velocity boundary condition (Case 1). The model with the highest geometrical complexity includes the details of the CRAH exit and the vane structure (Case 3). In order to have a representative momentum at the CRAH unit exit without the geometrical complexity of case 3, this work introduces another scenario with a narrow exit area (i.e., 16-inch width) along the CRAH length (Case 2). Since the detailed CRAH exit geometry in Case 3 is not valid for various CRAH units, Case 3 has been left out of the results in this study. It is noteworthy to mention that in case 2, CRAH airflow enters the domain with a 45-degree angle toward CA to have a similar directed flow due to the vane structure in case 3. However, the flow is downwards in Case 1. Fig. 7 illustrates the area that airflow exits from the

Table 2
Summary of model assumption.

IT Power	1000	KW
# of Racks	80	–
Rack Air Temperature Rise	14.6	°C
# of CRAH Units	10	–
Single CRAH Rated Flow Rate	8.02	m ³ /s
Single CRAH HX UA at the Rated Flow Rate	16.0	kW/K
CRAH Pressure Loss Factor, K_{cr}	141	–
Server Pressure Loss Factor, K_r	205	–
Perforated Tile Pressure Loss Factor, K_t	21	–
# of Perforated Tile in the Cold Aisle	120	–
# of Fans for iBP Configuration	120	–
# of Fans for fBP Configuration	40	–
Chiller Rated Cooling Capacity	1500	kW
Design COP of the Chiller	6.0	–
Condenser Water Temperature	35	°C

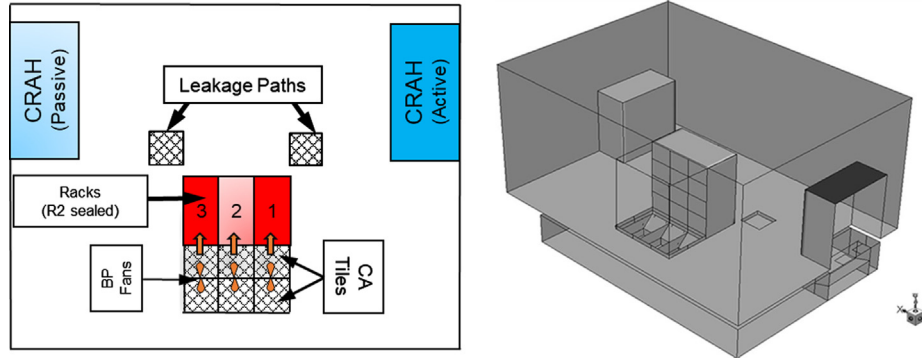


Fig. 6. Experimental facility layout (left) and 3-D CFD geometry (right).

CRAH unit in each of these three cases.

Fig. 8 shows the comparison of experimental and CFD temperatures for cases 1, 2 and 3, respectively. In this figure, the numbers in the legend correspond to the Φ values, and the word “L” indicates the leakage tile temperatures. $\Phi = 1.0$ means that the CRAH airflow rate is equal to that of racks. $\Phi = 1.34$ corresponds to the baseline configuration with no BP fans, where 1.0 unit goes to the racks, and the overhead of 0.34 leaks through the floor (~25% leakage). Also, root mean square error (RMSE) values are shown for three cases. The distribution of displayed data around 45 degree-line confirms that case 2 provides sufficient accuracy with minimal increase in the geometrical complexity. The numerically obtained temperatures for both case 2 and case 3 correlate favorably with measured temperatures considering the experimental uncertainty. However, the fully open case (case1) seems to over-predict the temperature non-uniformity across the rack inlets.

4. Results and discussion

The results section includes CFD-based optimization results of the baseline data center given in Section 2.3 with an emphasis on the effect of the CRAH exit boundary condition in parallel with the experimentally validated CFD. The subsequent parts examine how the optimum operating points and energy-saving potential of the CRAH BP method in various configurations deviate from that of the baseline QDC configuration. Parametric studies of various designs and operating conditions include different plenum heights, IT load density and server airflow rate.

4.1. CFD vs. TDM for the optimization of the baseline QDC

Fig. 9 presents the cooling power (W_c) and the associated chilled water temperatures (T_{chw}) provided by TDM and CFD, for both iBP and fBP configurations. In this figure, the solid and dashed lines represent the W_c and T_{chw} , respectively. Fig. 9 presents the impact of the CRAH exit flow area on the CFD-based optimization results for both CRAH exit

design cases 1 and 2 (i.e., full area downward flow and narrow opening with a directed flow). Previous studies [14,23] predicted higher temperature non-uniformities with the CRAH exit design case 1 at lower Φ values compared to the results by experimentally validated CFD model with CRAH exit design case 2.

In iBP and fBP configurations of EA with CRAH exit design case 2, the maximum differences for cooling power between the experimentally validated CFD and TDM, with well-mixed temperature and pressure assumptions, are 6.5% and 2.8%, respectively. At the optimal operating point, the difference is even lower than that. Due to the limited disagreement between the results, and less time and effort in the TDM approach, TDM may be a robust technique to do some practical optimization studies in EA data centers with similar designs.

On the other hand, CFD is the only option for the optimization of the CRAH BP for OA configuration, where the cooling system operates at lower evaporator temperatures because of higher temperature non-uniformities. TDM-based optimization of the CRAH BP method in the OA data center is not possible without any information about the temperature non-uniformity among the servers due to the highly non-uniform flow and temperature field in the data center. EA configuration limits non-uniformities to some extent, especially for Φ values greater or equal to 1.0, where TDM-based optimization is more reliable, at least for the baseline data center design discussed in this section. Parametric study scenarios in the next section considering various plenum heights, IT loads and server air flow rates will provide further limitations of TDM.

According to CFD results, in EA configuration, cooling power decreases 26% and 21% for iBP and fBP data centers, whereas OA exhibits a 21% reduction for both iBP and fBP data centers. It is important to note that the predicted optimal Φ value for EA may slightly vary depending on whether TDM or CFD is in use. CFD-based optimal results indicate an optimal Φ value of 0.9 for both iBP and fBP in the EA configuration. Even though the optimum value for the OA data center is $\Phi = 0.9$ in fBP configuration, it is between $0.7 \leq \Phi \leq 1.0$ for iBP configuration due to moderate fluctuations in this range.

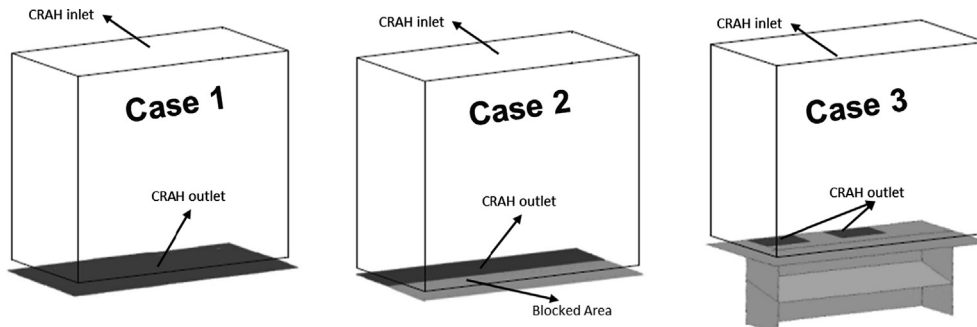


Fig. 7. CRAH exit boundary condition designs for three different cases.

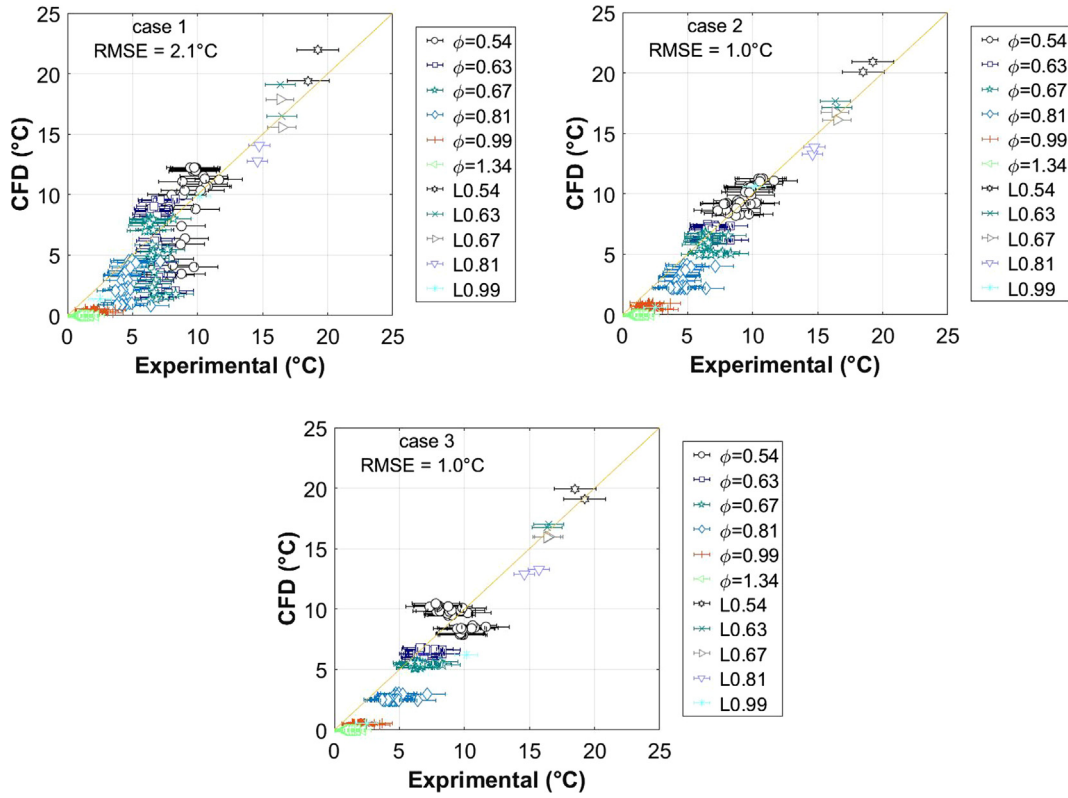


Fig. 8. Comparison of experimental and CFD temperature ($T - T_{cr,o}$) results for three cases.

4.2. Parametric analysis

Section 4.1 provided optimization results for the application of fBP and iBP in both EA and OA configurations, where QDC has a certain design. The parametric analysis is intended to study the impact of variations in the design of the data center, such as plenum height and IT configurations such as different server airflow rates and power levels on the optimization results in general.

4.2.1. Plenum height

Variation of plenum height can lead to changes in velocity and

pressure distribution inside the raised-floor volume. By raising the plenum height, the pressure variations reduce, and more uniform flow distribution appears as indicated in the literature [21]. This parametric study investigates three different plenum heights, including 36 (baseline), 28, and 20 in., and the impact of these scenarios on the CRAH BP method. Fig. 10 shows the optimization results for EA and OA data centers with iBP configuration (i.e., iEA and iOA, respectively) and fBP configuration (i.e., fEA and fOA, respectively). In this figure, the solid lines and dashed lines indicate the W_c on the primary axis and T_{chw} on the secondary axis, respectively.

For EA configurations (i.e., iEA and fEA) shown in Fig. 10, the

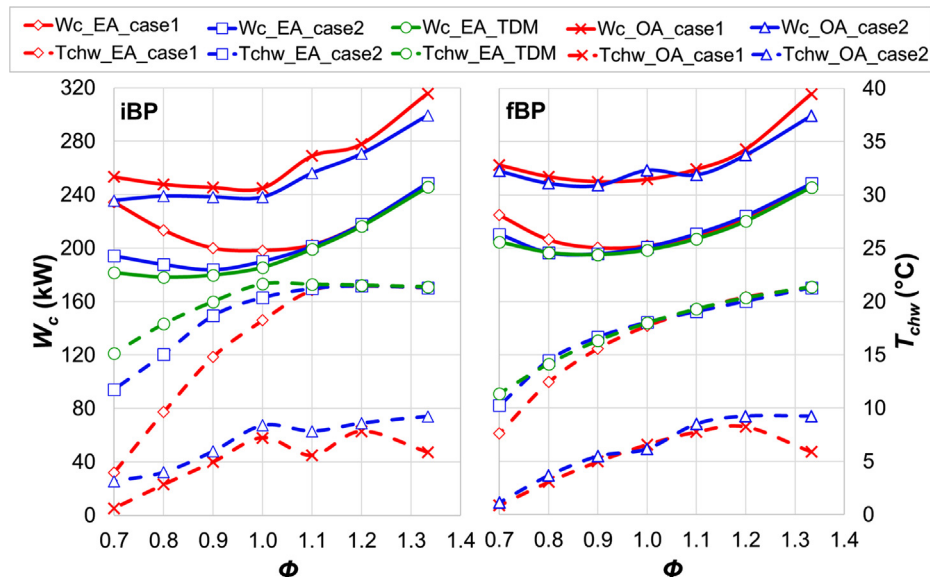


Fig. 9. Comparison of cooling power between CFD cases and TDM in iBP and fBP configuration.

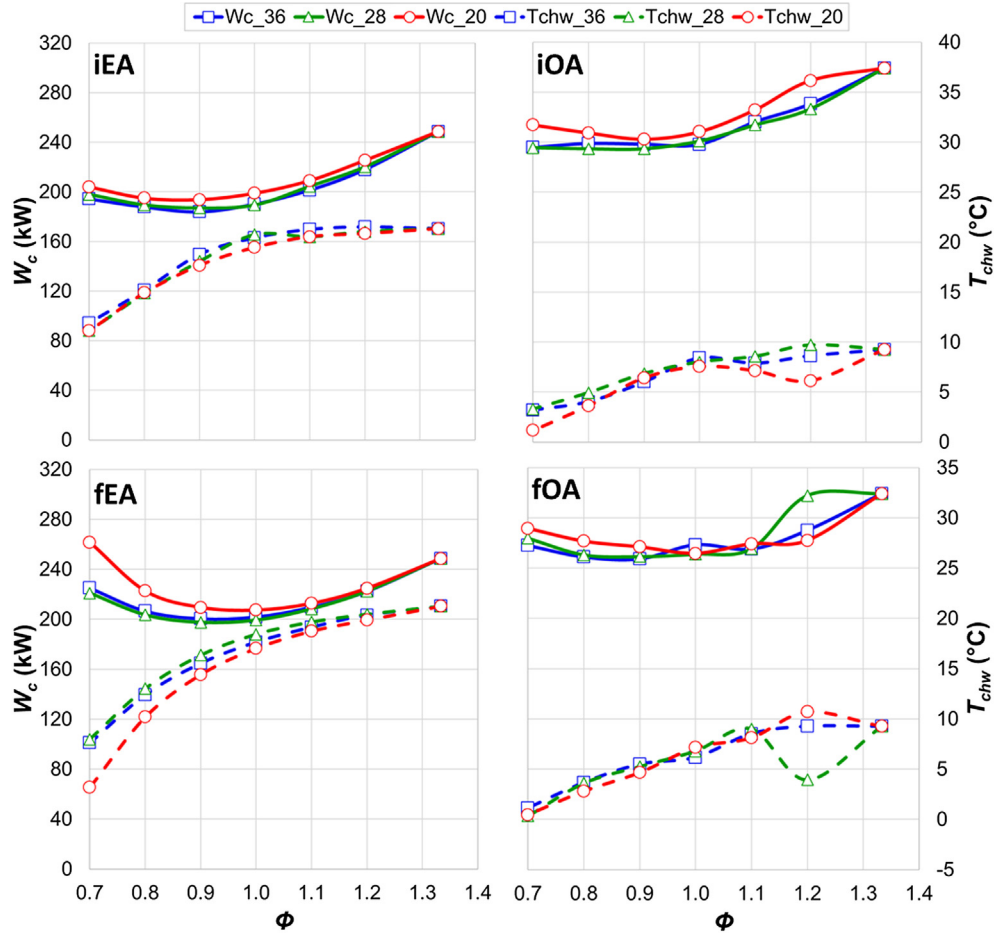


Fig. 10. Impact of plenum height on optimization results for various plenum heights in iEA, iOA, fEA, and fOA configurations.

estimated cooling power for plenum heights of 36-inch is just under the 28-inch, and the 20-inch scenario has the most cooling power due to the higher non-uniformity compared to others. Moreover, the EA CRAH BP case with a 20-inch plenum has less potential to reduce cooling power compared to others. The estimated cooling power for iOA follows a similar trend observed for iEA configuration. However, for fOA configuration, the cooling power predictions do not follow a predictable pattern.

For both iOA and iEA configurations, the optimum ϕ values occur at the vicinity of 0.9. In both fEA and fOA configurations, the optimum ϕ values take place about 0.9 for 36 and 28-inch plenum. However, it is around $\phi = 1.0$ for the 20-inch case for both fEA and fOA configurations. The error in the cooling power predictions by TDM compared to CFD is the highest in fEA configuration (22%), where CFD use is further justified.

4.2.2. Server airflow rate and IT load variations

This section initially discusses the impact of variation in the server airflow rate on the temperature non-uniformities and optimization results. As the server airflow rate decreases, the temperature difference across the racks (ΔT_r) increases, and there is a potential of increased temperature non-uniformity across the servers. Fig. 11 demonstrates the biggest difference of 11% between the cooling power estimated by CFD and TDM, which is a confirmation of increased non-uniformity. Hence, CRAH BP applications in data centers with higher ΔT_r and optimal ϕ values as low as 0.7 may require CFD analyses. On the other hand, there is a 4.2% error in the cooling power prediction by TDM at the optimal ϕ compared to the validated CFD. Considering the practicality of TDM, the ability to predict the optimal ϕ and the cooling

power at the optimal ϕ with reasonable accuracy, TDM can still be considered as a decent tool for parametric analyses and optimization.

Figs. 12 and 13 display the TDM-based optimization results of various levels of server airflow rate and IT load, respectively. In these figures, the solid and dashed lines indicate the cooling power (W_c) on the primary axis and the associated chilled water temperatures (T_{chw}) on the secondary axis, respectively. The baseline data center has a 1000 kW IT load and $\Delta T_r = 14.6$ °C.

The scenarios of various server airflow rates include $\Delta T_r = 14.6$ °C (baseline), 17.5 °C and 20.0 °C for 1000 KW IT load data center. CRAH

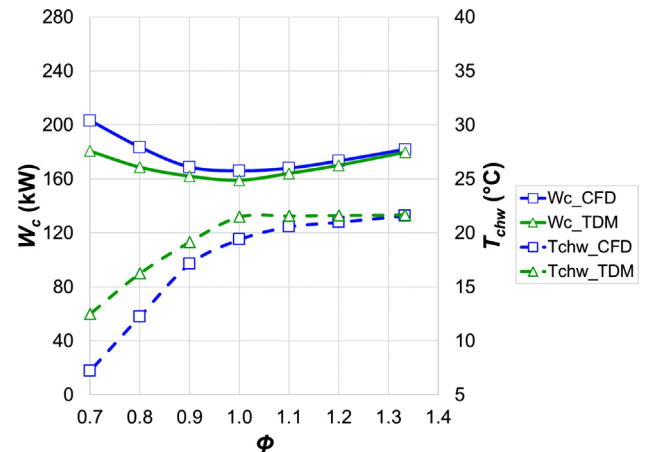


Fig. 11. Comparison of cooling power for $\Delta T_r = 20$ °C by TDM and CFD in iBP configuration.

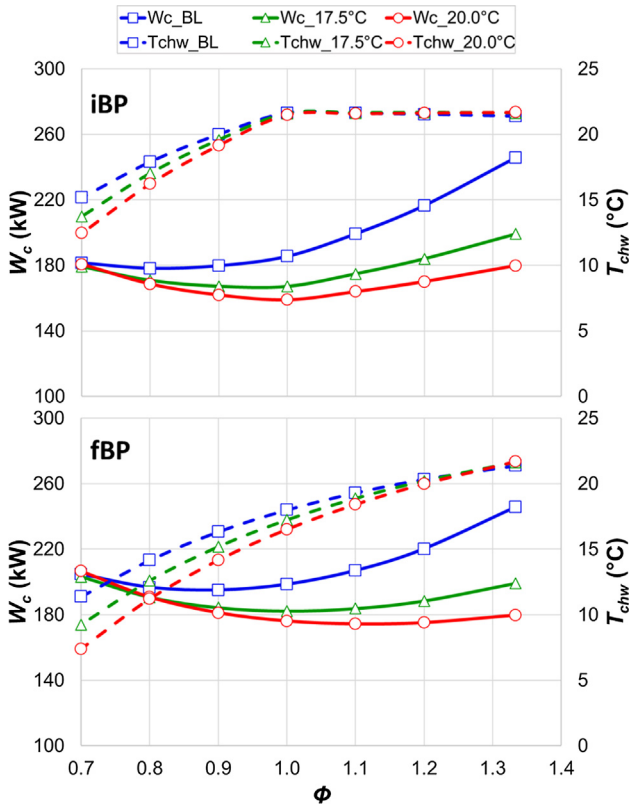


Fig. 12. Comparison of cooling power for various server airflow rates in iBP and fBP configuration.

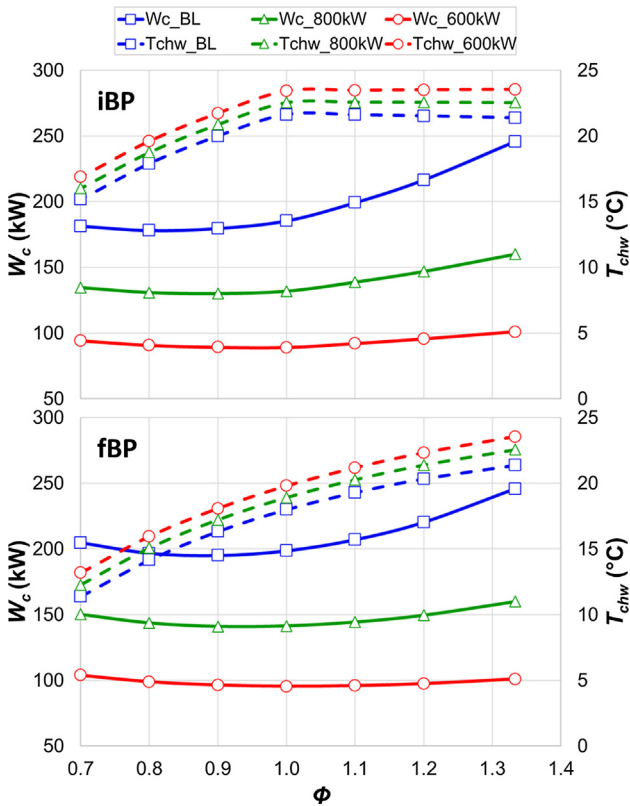


Fig. 13. Comparison of cooling power for various IT loads in iBP and fBP configurations.

BP method has lower energy saving potential in data centers with servers that have lower airflow rate (20.0 °C), where the cooling power decreases 11.5% and 3% for iBP and fBP, respectively. The optimal operating point for $\Delta T_r = 17.5$ °C scenario is around $\Phi = 1.0$ for both iBP and fBP configurations. However, in $\Delta T_r = 20$ °C scenario, it happens near 1.0 and 1.1 for iBP and fBP, respectively. Higher values of ΔT_r typically correspond to newer generations of IT equipment with more advanced thermal management, which confirms that legacy data centers are better candidates for CRAH BP application.

The scenarios of various IT load levels include 1000-kW (baseline), 800-kW and 600-kW in the same data center infrastructure for both iBP and fBP configurations. These scenarios can be considered as gradually increasing load levels in a data center over the years. The energy-saving potential is lower at lower load levels due to the oversized CRAH fan capacity and reduced fan power with reduced airflow requirements. As mentioned before, the CFD results of the baseline model demonstrate 26% and 21% reduction in cooling power for iBP and fBP configurations, respectively. However, these amounts are 11.7% and 5.5% in the 600-kW scenario, in which the CRAH BP method has the lowest benefit. The IT load level has a considerable impact on the optimum operating point of the CRAH BP method. The optimum Φ values in iBP configuration are 0.8, 0.9 and 1.0 for baseline, 800-kW and 600-kW cases, respectively. The optimum Φ values are less sensitive to the load variations in fBP configuration, where $\Phi = 0.9$ for the baseline and 800 kW cases and $\Phi = 1.0$ for the 600-kW case.

As indicated in the results, the energy-saving potential of the CRAH BP method depends on the efficiency of the baseline data center and the variations in the data center design and operation. For instance, at lower rates of floor leakage possible in tighter raised-floor designs, there is less benefit of using CRAH BP method. Variations in fan efficiencies, flow resistances, sizing of the cooling infrastructure equipment, the size and the number of BP fans likely lead to variations in the potential energy savings through the CRAH BP method. While reduced-order modeling tools in the literature allow testing some of these design variations, some of them may require more advanced modeling tools such as CFD. This study, as the first comprehensive investigation of the CRAH BP method using experimentally validated CFD modeling, determines where the CFD modeling can be avoided or needed. The energy analysis of the CRAH BP method requires holistic optimization of the data center IT and cooling infrastructure based on annual energy simulations using reduced-order modeling tools, provided that CFD confirms reasonable accuracy by these tools, especially at the optimal operation. Ultimately, the feasibility of the investment requires a thorough economic assessment.

5. Conclusion

The previous studies utilized reduced-order modeling tools (i.e., FNM and TDM) for the optimization of the CRAH BP method, which assumes uniform pressure and temperature at various zones in air-cooled EA data centers. Despite promising performance as a practical optimization tool, these tools ignore the impact of temperature and pressure non-uniformities on the optimization results, and they cannot predict the performance of the CRAH BP method in OA data centers with complex airflow patterns. Hence, CFD models are essential to investigate the CRAH BP method and evaluate the performance of TDM-based optimization tools. This study includes the experimental validation of a CFD model to evaluate the energy-saving potential of the CRAH BP method for iBP and fBP in both EA and OA configurations. Experimentally validated CFD results indicate instances (i.e., lower server airflow rates, shallow plenums) where reduced-order modeling tools exhibit higher errors in predicting optimal performance. This error is as high as 11% for $\Phi = 0.7$ and less than 4.2% at the optimal Φ for the baseline scenario. However, shallower plenums and decreased airflow rate requirements by the servers not only increase the temperature non-uniformities and but also decrease the potential for energy savings

with the CRAH BP method.

CRediT authorship contribution statement

Vahid Ebrahimpour Ahmadi: Methodology, Validation, Formal analysis, Investigation, Writing - original draft, Writing - review & editing, Visualization. **Hamza Salih Erden:** Conceptualization, Methodology, Resources, Writing - original draft, Writing - review & editing, Supervision, Project administration, Funding acquisition.

Declaration of Competing Interest

The authors declared that there is no conflict of interest.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.applthermaleng.2019.114685>.

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